

PHY 403 - MidSem-2 @ 11/10/25

Q Classical H for a charge particle in external EM field is

$$H_{\text{classical}} = \frac{1}{2m} (\mathbf{p} - q\mathbf{A})^2 + q\phi \quad \text{--- 0.5 marks}$$

$$\mathbf{B} = B_0 \hat{n} \Rightarrow \vec{A} = \frac{1}{2} (\vec{B} \times \vec{r}) \quad \text{with } \vec{B} = \nabla \times \vec{A}$$

For H-like atom

$$\therefore H_{\text{em}} = \frac{1}{2m} (\vec{p} + e\vec{A})^2 - \frac{Ze^2}{4\pi\epsilon_0 r} \quad \left. \vphantom{\frac{1}{2m}} \right\} \text{0.5 marks}$$

both or either: 0.5

$$\begin{aligned} &= \frac{p^2}{2m} + \frac{e}{2m} (\vec{p} \cdot \vec{A} + \vec{A} \cdot \vec{p}) + \frac{e^2 A^2}{2m} - \frac{Ze^2}{4\pi\epsilon_0 r} \\ &= -\frac{\hbar^2}{2m} \nabla^2 - \frac{Ze^2}{4\pi\epsilon_0 r} - \frac{i\hbar e}{2m} (\vec{\nabla} \cdot \vec{A} + \vec{A} \cdot \vec{\nabla}) + \frac{e^2 A^2}{2m} \end{aligned}$$

Now $\vec{\nabla} \cdot (\vec{A} \psi) = \vec{A} \cdot (\vec{\nabla} \psi) + \underbrace{(\vec{\nabla} \cdot \vec{A}) \psi}_{=0}$

0.5 marks

$$= \vec{A} \cdot (\vec{\nabla} \psi)$$

$$\therefore H_{\text{an}} = \frac{-\hbar^2}{2m} \nabla^2 - \frac{Ze^2}{4\pi\epsilon_0 r} - \frac{i\hbar e}{m} \vec{A} \cdot \vec{\nabla} + \frac{e^2 A^2}{2m}$$

$(\mu_B = \frac{e\hbar}{2m})$

$$H'_B = \frac{-i\hbar e}{m} \vec{\nabla} \cdot \vec{A} = -\frac{i\hbar e}{2m} (\vec{B} \times \vec{r}) \cdot \vec{\nabla} = \frac{e}{2m} \vec{B} \cdot \vec{L} = \mu_B \vec{L} / \hbar$$

$$\vec{M}_S = -g_S \mu_B \vec{S} / \hbar \Rightarrow H'_S = -\vec{M}_S \cdot \vec{B}$$

(spin)

$$\therefore H_{\text{EM}} = \frac{-\hbar^2}{2m} \nabla^2 - \frac{Ze^2}{4\pi\epsilon_0 r} + \frac{\mu_B}{\hbar} (\vec{L} + 2\vec{S}) \cdot \vec{B} + \frac{e^2 A^2}{2m}$$

0.5 mark

(1b) H with relativistic corrections (fine structure)

$$H = \frac{-\hbar^2}{2m} \nabla^2 - \frac{Ze^2}{4\pi\epsilon_0 r} + H'_1 + H'_2 + H'_3$$

1.5 marks: 0.5 for each

$$\left\{ \begin{array}{l} H'_1 = -C p^4 \quad ; \quad C \equiv \text{constant} \\ H'_2 = f(r) \vec{L} \cdot \vec{S} \quad (\text{spin-orbit}) \\ H'_3 = f(r) \nabla^2 \quad \text{or} \quad f(r) \delta(\vec{r}) \quad (\text{Darwin term}) \end{array} \right. \left\{ \begin{array}{l} \text{-ive can be in "C"} \end{array} \right.$$

$$H_{FS} = H'_1 + H'_2 + H'_3$$

$\therefore$ Using results from (1a)

$$H = H_{EM} + H_{FS}$$

$$= \frac{\hbar^2 \nabla^2}{2m} - \frac{Ze^2}{4\pi\epsilon_0 r} + \underbrace{\frac{\mu_B}{\hbar} (L_z + 2S_z) B_0}_{0.5 \text{ marks}} + \frac{e^2 A^2}{2m} + H_{FS}$$

$\therefore$ The complete equation is

$$H \psi(\mathbf{r}) = E \underbrace{\psi(\mathbf{r})}_{\text{total wavefunction}} \quad \left. \begin{array}{l} \rightarrow 1 \text{ mark} \\ 0.5 \text{ marks if } \psi(\vec{r}) \end{array} \right\}$$

$$\psi(\mathbf{r}) \equiv \psi(r) \chi_s$$

1c spin-orbit part is $H_2' = f(r) \vec{L} \cdot \vec{S}$

0.5 mark for
 For H-like atom, $\frac{e^2}{8m} (\vec{B} \times \vec{r})$ can be neglected in comparison with $(\vec{L} + 2\vec{S})$ term for typical lab $\vec{B}$ field (e.g. $B_{lab} < 10 \text{ T}$)

$$\frac{e^2}{8m} (\vec{B} \times \vec{r})^2 = \frac{e^2}{8m} B^2 r^2 \sin^2 \theta$$

$$\approx \frac{e^2 a_0^2 n^4 B^2}{8m Z^2}$$

$$\langle r^2 \rangle \approx a_0^2 \frac{n^4}{Z^2} \left\{ 1 + \dots \right\}$$

which

$$\frac{\mu_B}{\hbar} (L_z + 2S_z) B \approx \mu_B B \approx \frac{e \hbar B}{2m}$$

(h) not taken as $L_z \psi \approx m_s \hbar \psi$ & so

$$\therefore \frac{e^2 / 8m (B \times r)^2}{\vec{M} \cdot \vec{B}} \approx \frac{e a_0^2 n^4 B}{4 \hbar Z^2} \approx \frac{n^4}{Z^2} B \times 10^{-6}$$

Not needed, just for clarity

$$\therefore H = -\frac{\hbar^2}{2m} \nabla^2 - \frac{Ze^2}{4\pi\epsilon_0 r} + f(r) \vec{L} \cdot \vec{S} +$$

$$\frac{\mu_B}{\hbar} (L_z + 2S_z) B$$

Case I: Strong $\vec{B}$, $\vec{L} \cdot \vec{S}$ is not important
(perturbative effect) in 0th order

1 mark

$$\therefore \begin{cases} H_0 = -\frac{\hbar^2}{2m} \nabla^2 - \frac{Ze^2}{4\pi\epsilon_0 r} + \frac{\mu_B}{\hbar} (L_z + 2S_z) B \\ H' = f(r) \vec{L} \cdot \vec{S} \end{cases}$$

Case II: $\vec{L} \cdot \vec{S}$ is stronger compared to
external $\vec{M} \cdot \vec{B}$ effect ($L + 2S$ term)

1 marks

$$\begin{cases} H_0 = -\frac{\hbar^2}{2m} \nabla^2 - \frac{Ze^2}{4\pi\epsilon_0 r} + f(r) \vec{L} \cdot \vec{S} \\ H' = \frac{\mu_B}{\hbar} (L_z + 2S_z) B \end{cases}$$

Note: in the above, the deciding factor $\vec{L} \cdot \vec{S}$ is already mentioned. If not stated like this anywhere (0.5 mark is assigned for this)

(2a) For a 2-e system,
 $\alpha \equiv \uparrow$ (spin-up) ; $\beta \equiv \downarrow$ (spin-down)
 $1, 2 \equiv$ particle label (indistinguishable)

0.5 mark
 The four possibilities are (with $\vec{S} = \vec{S}_1 + \vec{S}_2$)

0.5 mark each
 $\uparrow\uparrow \quad \chi_{1,1} = \alpha(1)\alpha(2) \quad S=1 \quad M_S=1$

0.5 mark each
 $\uparrow\downarrow \quad \chi_{0,0} = \frac{1}{\sqrt{2}} \{ \alpha(1)\beta(2) - \beta(1)\alpha(2) \} \quad S=0, M_S=0$

0.5 mark each
 $\downarrow\uparrow \quad \chi_{1,0} = \frac{1}{\sqrt{2}} \{ \beta(1)\alpha(2) + \alpha(1)\beta(2) \} \quad S=1, M_S=0$

0.5 mark each
 $\downarrow\downarrow \quad \chi_{1,-1} = \beta(1)\beta(2) \quad S=1, M_S=-1$

where $\vec{S} = \vec{S}_1 + \vec{S}_2$ & $S^2 \chi = S(S+1) \chi$
 $S_z = S_{1z} + S_{2z}$ (in a.u.)

0.5 mark each if $\uparrow\downarrow = \alpha(1)\beta(2)$ & $\downarrow\uparrow = \beta(1)\alpha(2)$ } not sym.

0.5 marks { $\chi_{0,0} \equiv \text{spin singlet}$, $\chi_{1,0,\pm 1} \equiv \text{spin triplet}$
 0.5 marks { 0th energy level, degeneracy = 4 (E : no dependence on $\vec{S}$)

(2b) The complete wavefunction is multiplication of spatial $\psi(r)$ and spin χ part

$$\psi(q) = \psi(r) \chi(s)$$

Now $\psi(r)$ can be symmetric or anti-symmetric

$$\therefore \psi(\vec{r}) = \psi_{\pm}(\vec{r})$$

$$\chi = \chi_{\pm} \quad (\text{from 2a})$$

Since total wavefunction must be anti-symmetric (Pauli's Exclusion Principle)

0.5 marks {
$$\psi(q_1, q_2) = \begin{cases} \psi_+(\vec{r}_1, \vec{r}_2) \chi_- & ; \text{ para state} \\ \psi_-(\vec{r}_1, \vec{r}_2) \chi_+ & ; \text{ ortho state} \end{cases}$$

$$\psi_{\pm} = \frac{1}{\sqrt{2}} \{ \psi_1(\vec{r}_1) \psi_2(\vec{r}_2) \pm \psi_2(\vec{r}_1) \psi_1(\vec{r}_2) \}$$

$$\text{or } \psi_{\pm}(\vec{r}_1, \vec{r}_2) = \frac{1}{\sqrt{2}} \left\{ \psi_{n_1 l_1 m_1}(\vec{r}_1) \psi_{n_2 l_2 m_2}(\vec{r}_2) \pm \psi_{n_2 l_2 m_2}(\vec{r}_1) \psi_{n_1 l_1 m_1}(\vec{r}_2) \right\}$$

& 0.5 marks for $\psi_{\pm}$ expression

(2c) The spatial part of wavefunction for 1- e^- in ground and other in excited state is

$$\psi(\vec{r}_1) \equiv \text{ground} = \psi_{100}(\vec{r}_1)$$

$$\psi(\vec{r}_2) \equiv \text{excited } (n \geq 2) = \psi_{n l m}(\vec{r}_2)$$

The proper symmetrized ψ is

$$\psi_{\pm} = \frac{1}{\sqrt{2}} \left\{ \psi_{100}(\vec{r}_1) \psi_{n l m}(\vec{r}_2) \pm \psi_{n l m}(\vec{r}_1) \psi_{100}(\vec{r}_2) \right\}$$

$$E^{(1)} = \langle \psi_{\pm} | H' | \psi_{\pm} \rangle = J_{nl} \pm K_{nl}$$

$$H' = \frac{1}{r_{12}}$$

$$K_{nl} = \int \psi_{1\omega}^*(\vec{r}_1) \psi_{nlm}^*(\vec{r}_2) \frac{1}{r_{12}} \psi_{nlm}(\vec{r}_1) \psi_{1\omega}(\vec{r}_2) d\vec{r}_1 d\vec{r}_2$$

$K_{nl} \equiv$ exchange term

$$E = E^{(0)} + E^{(1)}$$

$$= -\frac{Z^2}{2} \left(1 + \frac{1}{n^2}\right) + J_{nl} \pm K_{nl}$$

both J_{nl} & K_{nl} are positive (in general)

$$E_{nl}^{(1)} = J_{nl} \pm K_{nl}$$

↓ mark ← {

$$= J_{nl} - \frac{1}{2} (1 + 4 \vec{S}_1 \cdot \vec{S}_2) K_{nl}$$

$$= J_{nl} - \frac{1}{2} (1 + \vec{\sigma}_1 \cdot \vec{\sigma}_2) K_{nl}$$

where $\vec{S}_i = \frac{1}{2} \vec{\sigma}_i$

1 $\vec{S} = \vec{S}_1 + \vec{S}_2$

with $|\vec{S}_1| = |\vec{S}_2| = \frac{1}{2}$
(in a.u.)

& $|\vec{S}| = \begin{cases} 0 \\ 1 \end{cases}$

$$S^2 = S_1^2 + S_2^2 + 2\vec{S}_1 \cdot \vec{S}_2$$

1- mark

$$\Rightarrow \vec{S}_1 \cdot \vec{S}_2 = \frac{1}{2} S^2 - \frac{3}{4}$$

$$= \begin{cases} -3/4 & ; S=0 \\ 1/4 & ; S=1 \end{cases}$$

$$\therefore E_m^{(1)} = \begin{cases} J_{nl} + K_{nl} & ; S=0 \\ J_{nl} - K_{nl} & ; S=1 \end{cases}$$

* Ortho (Ψ_-) has lower energy than para (Ψ_+) state

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